Unit 1: Technology, IP and Security

Data privacy and Network and security: 50 multiple-choice questions with answers

Name: _____ **Class:** _____ **Score:** _____ / 50

Choose **one** answer (A, B, C or D) for every question and circle it. The answer key and short explanations are on the last pages. Questions follow Sprint 1.4 (Data privacy) and Sprint 1.5 (Network and security). The answer key names the lesson for each question.

Part 1: Data privacy (Sprint 1.4)

1. Which best describes "automated data collection"?

A Collecting data automatically, often without users' full knowledge or consent
B A method where users manually approve every single data point collected
C A process that only happens with written government permission
D Automatically deleting all of a user's personal data after every single browsing session ends

2. Which organisation was accused in 2018 of harvesting data from up to 87 million Facebook users without their consent?

A Yahoo
B Microsoft
C The Network Advertising Initiative
D Cambridge Analytica

3. How did Cambridge Analytica initially obtain much of its data?

A By covertly hacking into several different national government servers
B Through a public records request
C Through a personality-quiz app connected to Facebook's API
D From a leaked Yahoo database

4. What is "profiling" in a data-privacy context?

A Verifying someone's identity by comparing a submitted photo ID against a government database
B Analysing personal data to categorise or predict someone's behaviour and preferences
C Encrypting a user's personal files
D Deleting inactive accounts

5. Which of these is an example of profiling described in the material?

A Receiving a routine software update notification
B Browsing for a t-shirt online and then seeing more shirt ads afterward
C Getting a two-factor authentication code by text
D A teacher manually grading an exam and recording the score in a paper gradebook

6. What is one privacy concern with surveillance cameras in stores and on roads?

A People are often unaware their behaviour is being recorded and analysed
B They cannot legally be installed without a special government permit first
C They only work during business hours
D They require the subject's written consent every time

7. Why does data privacy matter for individual freedom, according to the material?

A It has no connection to personal freedom
B It only benefits large corporations rather than ordinary individual people
C It is solely a legal formality with no real impact
D It lets people control what personal information they share and with whom

8. What challenge do differing global privacy laws create for international companies?

- **A** None — all countries follow the exact same rules
- **B** Companies are completely banned from collecting any personal data worldwide
- **C** Conflicting compliance and enforcement requirements across regions
- **D** It eliminates the need for privacy policies

9. Which of the following is listed as a POSITIVE impact of data use?

- **A** Identity theft caused by a major corporate data breach
- **B** Discrimination and bias
- **C** Enhanced personalisation of products and services
- **D** Widening the digital divide

10. Which of the following is listed as a NEGATIVE impact of data misuse?

- **A** Improved decision-making
- **B** Vulnerability to fraud and identity theft
- **C** Steady long-term economic growth for the whole country
- **D** Better public services

11. What does “digital divide” refer to in this context?

- **A** Growing inequity for people who lack access to, or literacy with, digital tools
- **B** A disagreement between two tech companies
- **C** A specialised type of military-grade data-encryption algorithm used by major banks
- **D** A privacy law passed in one country

12. What is a data breach?

- **A** A routine scheduled software update that every device installs automatically overnight
- **B** A backup copy of a database
- **C** A new privacy regulation
- **D** Unauthorised access to, sharing of, or theft of sensitive or protected information

13. Which company’s breach (occurring 2013–2016) exposed data from over 3 billion accounts?

- **A** Microsoft
- **B** Cambridge Analytica
- **C** A retail surveillance company
- **D** Yahoo

14. The Microsoft Exchange email server attack affected roughly how many US organisations?

- **A** Exactly 100 large multinational technology corporations only, worldwide
- **B** No real organisations
- **C** About 30,000, from small businesses to local governments
- **D** Only individual home users

15. In the Activity, someone creates a fake Facebook account using Eric’s name and photo. This is best described as:

- **A** A completely standard and expected security feature on any social platform
- **B** A form of identity theft / impersonation
- **C** A type of two-factor authentication
- **D** A coincidence with no privacy impact

16. According to the suggested solution, what should Eric do first about the fake account?

- **A** Report the fake account to Facebook
- **B** Delete his real account
- **C** Give the impersonator his login details to “prove” he's real
- **D** Ignore it — it will resolve itself

17. What is recommended when you must use sensitive services (like banking) on public Wi-Fi?

A Use a Virtual Private Network (VPN) to encrypt your connection
B Turn off your phone's lock screen
C Share the public Wi-Fi password with as many other people as possible
D Disable your antivirus software

18. In Challenge 1, which feature helps stop unauthorised people from accessing Ms. Anna's new phone?

A Uninstalling every single app currently installed on the phone
B Turning on location services for all apps
C Disabling all passwords for convenience
D Screen lock (PIN, password, pattern, or biometric)

19. In Challenge 2, how can John reduce the targeted ads he keeps seeing?

A Buy more items so the algorithm "gives up"
B Post even more of his personal data publicly across every social media platform
C Adjust ad-personalisation settings and use privacy tools like ad blockers
D Permanently disconnect his phone from the internet

20. In Challenge 3, what is the first step recommended for Sophia to audit her social media privacy?

A Immediately and permanently delete every single one of her social media accounts
B List all the platforms she's active on and review each one's privacy settings
C Make every post public so she can "see what happens"
D Share her passwords with close friends for backup

21. Which one of these statements is TRUE?

A The people whose data Cambridge Analytica used had all given full, informed consent for that use.
B Automated data collection can happen without users being aware of it or giving explicit consent.
C Profiling is used only for harmful purposes and has no legitimate applications.
D Because surveillance footage is often anonymous, it can never be used to identify or target specific individuals.

22. Which one of these statements is TRUE?

A Data privacy laws and regulations differ from country to country based on social, economic, and cultural factors.
B Profiling is used only for harmful purposes and has no legitimate applications.
C Because surveillance footage is often anonymous, it can never be used to identify or target specific individuals.
D Using the same simple password for every account is a recommended way to protect your privacy.

23. Which one of these statements is TRUE?

A Because surveillance footage is often anonymous, it can never be used to identify or target specific individuals.
B Using the same simple password for every account is a recommended way to protect your privacy.
C Regularly updating your devices and apps has no real effect on protecting your privacy or security.
D Government use of data for public-safety purposes is mentioned as one of the positive impacts of data use.

24. Which one of these statements is TRUE?

A Using the same simple password for every account is a recommended way to protect your privacy.

B Regularly updating your devices and apps has no real effect on protecting your privacy or security.

C The Yahoo breach (2013–2016) is described as one of the largest data breaches in history.

D The people whose data Cambridge Analytica used had all given full, informed consent for that use.

25. Which one of these statements is TRUE?

A Regularly updating your devices and apps has no real effect on protecting your privacy or security.

B The people whose data Cambridge Analytica used had all given full, informed consent for that use.

C Enabling two-factor (multi-factor) authentication adds an extra layer of security beyond a password alone.

D Profiling is used only for harmful purposes and has no legitimate applications.

Part 2: Network and security (Sprint 1.5)

26. What is the primary definition of a cyber-attack?

A A scheduled operating system update run across a company network

B An attempt to gain unauthorised access to a device or its data

C An automated cloud backup that archives user databases remotely

D A hardware installation that upgrades local network router speeds

27. Which type of devices did the 2016 Mirai botnet attack specifically target?

A Internet of Things (IoT) devices such as cameras and routers

B Cloud data-centre servers running optical storage arrays

C Mainframe supercomputers inside university research labs

D Satellite ground-station antennas sending military signals

28. Which type of cyber attack overwhelms a server or website with massive traffic so it becomes unavailable?

A A Multi-Factor Authentication check across several nodes

B A database encryption and decryption routine run on a schedule

C A local file compression and archive backup process daily

D A Denial of Service (DoS) attack that floods the website

29. Which term describes a network of compromised devices controlled remotely by cybercriminals?

A A virtual private network set up for encrypted remote working

B A local area network switch with a routing table to manage

C A domain name system root server directory for the internet

D A botnet, a group of hijacked devices under remote control

30. Which type of malware spreads by itself across network vulnerabilities without any human interaction?

A Screen-locker ransomware showing fake police payment demands

B Keylogger spyware that records every key pressed on a keyboard

C A worm, which copies itself across a network without any help

D Adware that shows pop-up adverts inside free web applications

31. What is data encryption?

- **A** Deleting old temporary cache files from a hard disk automatically
- **B** Scrambling data into an unreadable format that needs a key to read
- **C** Sending plain text files over a public wireless connection
- **D** Installing a hardware token that makes one-time password strings

32. What acts as a protective shield that blocks untrustworthy sources from reaching a private network?

- **A** A firewall, which filters traffic entering the network
- **B** A domain name system directory that resolves website names
- **C** A USB hardware key used to sign in to online accounts
- **D** A public cloud storage repository used for data backups

33. Why was the Mirai botnet harder to defend against than a standard DoS attack?

- **A** Its bots sent the attack traffic from many different places at once
- **B** It used quantum decryption keys that bypassed every hardware firewall
- **C** It erased domain name system routing tables across the global internet
- **D** It only affected offline desktop computers cut off from their routers

34. Which psychological tactics do phishing emails commonly use to manipulate victims?

- **A** Technical jargon, formal code syntax and cryptographic formulas
- **B** Device diagnostic logs, system error codes and driver release notes
- **C** Academic citations, study abstracts and library metadata tags
- **D** Urgency, greed and fear that push people to click without thinking

35. How does a Trojan horse malware trick users into installing it?

- **A** It spreads through open network ports with no user involvement
- **B** It sends huge numbers of ping requests to public DNS servers
- **C** It disguises itself as legitimate software that looks useful
- **D** It locks the phone screen as soon as a text message arrives

36. Which type of ransomware encrypts a victim's data and demands payment for the decryption key?

- **A** A screen locker that freezes the display but leaves files untouched
- **B** Crypt ransomware, which locks the files themselves with encryption
- **C** Scareware pop-ups that show fake alerts to sell antivirus plans
- **D** An adware script that redirects browser tabs to shopping sites

37. Which type of malware gives attackers administrative access while hiding other malicious code?

- **A** A keylogger built only to capture what is typed on a keyboard
- **B** A rootkit, which hides itself and other malware on the system
- **C** An adware extension showing banner adverts on web pages
- **D** A screen-locker app that blocks the display until a ransom is paid

38. What is the main security purpose of keeping software regularly updated?

- **A** Fixing bugs and patching known security vulnerabilities in programs
- **B** Deleting user profiles and resetting browser bookmarks automatically
- **C** Raising the RAM capacity and processor speed of the hardware
- **D** Turning local text files into encrypted public databases

39. Which verification method uses hardware or software to generate short-lived One-Time Passwords (OTPs)?

- **A** A fingerprint scanner built into the phone's hardware chip
- **B** A pattern lock drawn on a nine-point touch screen grid of dots
- **C** A four-digit Personal Identification Number (PIN) code
- **D** Security tokens that produce a new code every short period

40. In the 2016 Mirai attack, how did the creators make money from their botnet?

- **A** By demanding Bitcoin ransoms to decrypt victims' hard drives
- **B** By stealing card details from IP cameras to sell on dark web markets
- **C** By selling anti-DDoS protection to the firms they attacked
- **D** By selling smart home appliances directly to online shoppers

41. An attacker captures confidential passwords by recording every character typed on a keyboard. Which malware was used?

- **A** Double extortion ransomware that exports data to public servers
- **B** Adware extension that generates commercial pop-up adverts
- **C** Keylogger spyware that logs every key pressed on the keyboard
- **D** A self-spreading worm that exploits server vulnerabilities

42. In the Facebook data breach linked to the "View As" feature, how did attackers obtain millions of user profiles without traditional malware?

- **A** By intercepting unencrypted Wi-Fi traffic at public internet cafes
- **B** By exploiting a flaw in the "View As" feature to steal access tokens
- **C** By deploying screen-locker ransomware across corporate servers
- **D** By brute-forcing four-digit PIN codes on users' mobile phones

43. In the 2010 Stuxnet attack, how did the worm get into the isolated industrial network of the targeted Iranian nuclear facility?

- **A** Through infected USB drives carried into the air-gapped network
- **B** Through phishing emails sent to power plant administrators
- **C** By launching a huge DDoS attack against DNS servers
- **D** By downloading fake software updates from unofficial websites

44. A school uses strict web filtering to block inappropriate content, but it also blocks legitimate educational research sites. Which cybersecurity trade-off does this show?

- **A** Accessibility versus multi-factor authentication tools
- **B** Data encryption versus the ongoing cost of storage
- **C** Privacy versus video surveillance monitoring systems
- **D** Openness versus control over what users can reach online

45. How does Double Extortion Ransomware differ from standard Crypt Ransomware?

- **A** It only locks the display screen and leaves local files unchanged
- **B** It shows fake antivirus warnings and demands small service fees
- **C** It uses security tokens to create one-time password strings
- **D** It encrypts the data and also exports it, threatening to sell it

46. Which one of these statements is TRUE?

A IoT devices such as smart cameras and routers are often targeted by botnets because they have strong, unchangeable default security controls.

B Phishing attacks rely only on hardware cable faults, not on psychological tactics such as urgency or fear.

C A Denial of Service (DoS) attack typically comes from a single source and is easier to defend against than a DDoS attack.

D Adware is a type of malware that restricts system access and demands ransom payments to decrypt files.

47. Which one of these statements is TRUE?

A Phishing attacks rely only on hardware cable faults, not on psychological tactics such as urgency or fear.

B Antivirus software uses a database of known definitions to stop unauthorised programs from harming a computer.

C Adware is a type of malware that restricts system access and demands ransom payments to decrypt files.

D The 2017 NotPetya ransomware attack was designed purely for financial gain, so every victim could reliably recover their files by paying the ransom.

48. Which one of these statements is TRUE?

A In a Distributed Denial of Service (DDoS) attack, compromised devices called "bots" form a network known as a botnet.

B Adware is a type of malware that restricts system access and demands ransom payments to decrypt files.

C The 2017 NotPetya ransomware attack was designed purely for financial gain, so every victim could reliably recover their files by paying the ransom.

D IoT devices such as smart cameras and routers are often targeted by botnets because they have strong, unchangeable default security controls.

49. Which one of these statements is TRUE?

A Multi-factor authentication (MFA) protects devices by requiring two or more verification methods before access is granted.

B The 2017 NotPetya ransomware attack was designed purely for financial gain, so every victim could reliably recover their files by paying the ransom.

C IoT devices such as smart cameras and routers are often targeted by botnets because they have strong, unchangeable default security controls.

D Phishing attacks rely only on hardware cable faults, not on psychological tactics such as urgency or fear.

50. Which one of these statements is TRUE?

A IoT devices such as smart cameras and routers are often targeted by botnets because they have strong, unchangeable default security controls.

B Phishing attacks rely only on hardware cable faults, not on psychological tactics such as urgency or fear.

C Adware is a type of malware that restricts system access and demands ransom payments to decrypt files.

D High levels of system security can sometimes reduce user accessibility, creating a trade-off that organisations must balance.

Answer key

Teacher and student copy.

No.	Answer	Lesson	Why
1	**A** Collecting data automatically, often without users' full knowledge or consent	Automated Data Collection	Automated data collection can happen quietly in the background — for example, apps mining account activity or cameras analysing shopper behaviour — often without people realising it.
2	**D** Cambridge Analytica	Cambridge Analytica Case	Cambridge Analytica combined this data with voter information and psychological profiling to target political ads and influence public opinion.
3	**C** Through a personality-quiz app connected to Facebook's API	Cambridge Analytica Case	The app could access data from people who took the quiz and, through the same permission, data about their friends too — even without the friends' own consent.
4	**B** Analysing personal data to categorise or predict someone's behaviour and preferences	Profiling & Surveillance	Profiling uses information such as browsing history, purchases, and demographics to make predictions — it's behind things like targeted ads and credit scoring.
5	**B** Browsing for a t-shirt online and then seeing more shirt ads afterward	Profiling & Surveillance	This is a common result of profiling: past browsing and purchase behaviour shapes the ads a person is shown later.
6	**A** People are often unaware their behaviour is being recorded and analysed	Profiling & Surveillance	Even when footage seems anonymous, video analytics can still connect it to habits, appearances, and other identifying details.
7	**D** It lets people control what personal information they share and with whom	Legal, Social & Economic Implications	Privacy lets people set boundaries around their information; without it, trust in relationships and institutions can break down.
8	**C** Conflicting compliance and enforcement requirements across regions	Legal, Social & Economic Implications	Some countries regulate data collection tightly while others favour open access, creating incompatible standards for companies operating across borders.
9	**C** Enhanced personalisation of products and services	Ethical Perspectives (Pros & Cons)	Data lets companies understand customer needs closely enough to build more personalised, satisfying products and services.
10	**B** Vulnerability to fraud and identity theft	Ethical Perspectives (Pros & Cons)	Sensitive data that is poorly managed can be exploited for fraud, identity theft, and manipulation.
11	**A** Growing inequity for people who lack access to, or literacy with, digital tools	Ethical Perspectives (Pros & Cons)	As more opportunities depend on data and digital access, people without that access or know-how risk being left further behind.

No.	Answer	Lesson	Why
12	**D** Unauthorised access to, sharing of, or theft of sensitive or protected information	Data Breaches	Breaches can happen through cyberattacks, theft, or simple human error, and can expose personal information on a massive scale.
13	**D** Yahoo	Data Breaches	The exposed data included names, emails, phone numbers, birth dates, passwords, and security questions.
14	**C** About 30,000, from small businesses to local governments	Data Breaches	The attack gave hackers access to email accounts across a wide range of organisations, not just large enterprises.
15	**B** A form of identity theft / impersonation	Real-World Scenarios	The impersonator could deceive Eric's contacts and misuse his identity, which is a serious privacy and reputational harm.
16	**A** Report the fake account to Facebook	Real-World Scenarios	Reporting the impersonating account is the first recommended step, along with warning friends and reviewing his own privacy settings.
17	**A** Use a Virtual Private Network (VPN) to encrypt your connection	Protecting Your Privacy	Public Wi-Fi is vulnerable to hacking, and a VPN encrypts your traffic so it's harder to intercept.
18	**D** Screen lock (PIN, password, pattern, or biometric)	Real-World Scenarios	A secure screen lock is one of the most basic and effective ways to prevent unauthorised access to a device.
19	**C** Adjust ad-personalisation settings and use privacy tools like ad blockers	Real-World Scenarios	Turning off ad personalisation where available, using ad blockers, and clearing cookies all help reduce targeted advertising.
20	**B** List all the platforms she’s active on and review each one’s privacy settings	Real-World Scenarios	Auditing starts with knowing exactly which platforms you use, then reviewing each one’s privacy policy and settings.
21	**B** Automated data collection can happen without users being aware of it or giving explicit consent.	True or false review	Examples include social platforms mining account data quietly and surveillance cameras analysing behaviour without direct notice.
22	**A** Data privacy laws and regulations differ from country to country based on social, economic, and cultural factors.	True or false review	Some countries regulate data collection tightly to protect privacy, while others favour open data access to encourage innovation.
23	**D** Government use of data for public-safety purposes is mentioned as one of the positive impacts of data use.	True or false review	Improved crime prevention, emergency response, and security outcomes are listed as potential benefits, alongside the risks of excessive surveillance.
24	**C** The Yahoo breach (2013–2016) is described as one of the largest data breaches in history.	True or false review	It affected over 3 billion accounts, exposing details like names, emails, phone numbers, and passwords.
25	**C** Enabling two-factor (multi-factor) authentication adds an extra layer of security beyond a password alone.	True or false review	It requires a second form of verification, such as a code sent to your phone, making accounts harder to break into.

No.	Answer	Lesson	Why
26	**B** An attempt to gain unauthorised access to a device or its data	Cyber-attacks	A cyber-attack is an attempt to gain unauthorised access to devices or to retrieve data from them.
27	**A** Internet of Things (IoT) devices such as cameras and routers	Botnets	Mirai infected poorly secured IoT devices such as smart cameras and routers.
28	**D** A Denial of Service (DoS) attack that floods the website	Denial of Service	A Denial of Service attack floods a server or website with traffic until it cannot respond to real users.
29	**D** A botnet, a group of hijacked devices under remote control	Botnets	A botnet is a network of compromised devices ('bots') controlled remotely by attackers.
30	**C** A worm, which copies itself across a network without any help	Malware	A worm self-propagates through network vulnerabilities and needs no user action to spread.
31	**B** Scrambling data into an unreadable format that needs a key to read	Encryption	Encryption encodes data into an unreadable format so only someone with the right key can read it.
32	**A** A firewall, which filters traffic entering the network	Network Defence	A firewall filters traffic and blocks untrusted sources from accessing a private network.
33	**A** Its bots sent the attack traffic from many different places at once	Botnets	Mirai was a distributed attack: traffic came from huge numbers of bots in many locations, so it could not simply be blocked at one source.
34	**D** Urgency, greed and fear that push people to click without thinking	Phishing	Phishing messages play on emotions such as urgency, greed and fear to trick users into clicking or sharing details.
35	**C** It disguises itself as legitimate software that looks useful	Malware	A Trojan pretends to be genuine software, so users install it themselves.
36	**B** Crypt ransomware, which locks the files themselves with encryption	Ransomware	Crypt ransomware encrypts the victim's files and demands payment for the key needed to decrypt them.
37	**B** A rootkit, which hides itself and other malware on the system	Malware	A rootkit gives attackers administrator-level access and conceals other malicious code from the user and security tools.
38	**A** Fixing bugs and patching known security vulnerabilities in programs	Network Defence	Updates fix bugs and patch security holes that attackers could otherwise exploit.
39	**D** Security tokens that produce a new code every short period	Authentication	Security tokens (hardware or software) generate short-lived one-time passwords.
40	**C** By selling anti-DDoS protection to the firms they attacked	Botnets	The creators ran a business selling anti-DDoS protection, profiting from the very attacks they launched.
41	**C** Keylogger spyware that logs every key pressed on the keyboard	Malware	A keylogger is spyware that records keystrokes, letting attackers capture passwords as they are typed.
42	**B** By exploiting a flaw in the "View As" feature to steal access tokens	Data Breaches	Attackers exploited a flaw in the "View As" feature to steal access tokens, which let them take over profiles with no malware.

No.	Answer	Lesson	Why
43	**A** Through infected USB drives carried into the air-gapped network	Stuxnet	The network was air-gapped, so the worm was carried in on infected USB drives.
44	**D** Openness versus control over what users can reach online	Security Trade-offs	Tighter control keeps users safe but reduces open access to useful educational content: openness versus control.
45	**D** It encrypts the data and also exports it, threatening to sell it	Ransomware	Double extortion ransomware both encrypts the data and steals (exports) it, threatening to sell or leak it if the ransom isn't paid.
46	**C** A Denial of Service (DoS) attack typically comes from a single source and is easier to defend against than a DDoS attack.	True or false review	A DoS attack uses fewer IP addresses from one source, whereas a DDoS attack comes from many distributed bots.
47	**B** Antivirus software uses a database of known definitions to stop unauthorised programs from harming a computer.	True or false review	Antivirus software compares programs against a database of known malware definitions.
48	**A** In a Distributed Denial of Service (DDoS) attack, compromised devices called "bots" form a network known as a botnet.	True or false review	The bots together form a botnet, which attackers use to flood a target with traffic.
49	**A** Multi-factor authentication (MFA) protects devices by requiring two or more verification methods before access is granted.	True or false review	MFA asks for two or more different proofs of identity, so a stolen password alone is not enough.
50	**D** High levels of system security can sometimes reduce user accessibility, creating a trade-off that organisations must balance.	True or false review	Stronger security controls can make systems harder to use, so organisations balance accessibility against security.